

AE460B Spacecraft Design

TM-02: Electrical Power System Design

Manifest Destiny

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This technical memorandum presents the preliminary design and electrical analysis of the spacecraft Electrical Power Subsystem (EPS) developed for the Solar Gravitational Lens (SGL) mission. The mission will travel beyond 650 AU. Due to this immense distance from the sun, a radioisotope-based design is selected, consisting of a Multi-Hundred Watt Radioisotope Thermoelectric Generator (MHW-RTG) with a regulated 28 V direct energy transfer bus. It will also have Nickel-Hydrogen battery storage, with battery sizing performed to ensure margin for transient loads and correct voltage distribution. The power and mission modes are defined, and contingency plans are introduced for the nominal and peak loads of 78 Watts and 98 Watts. The RTG degradation is accounted for in the energy balance over the 35-year duration. Results show the energy margin as positive during all mission modes, with a substantial depth-of-discharge margin for peak events. The proposed design satisfies the requirements for mission reliability, longevity, and stability for deep-space operation.

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I. Mission and Power Requirements

The electrical power systems on The SGL Mission are designed for an almost entirely deep-space mission, meaning the relevancy of the sun becomes rendered to nothing as the space craft approaches its operating point. The subsystems which require power for this mission are propulsion, On-Board Computer (OBC), Telemetry Tracking and Control (TT&C), Payload, Mechanical, Thermal and ADCS. This will be a low-power mission with an average power requirement of 78 Watts, with peak power at 98 Watts during the travel to the operating point. This will be an eclipse-free travel, with power systems mainly focused in deep-space, there is no relevancy for switched modes, although batteries will be implemented as a contingency for grid outages and power overload.

A. Mission Scenario

Our mission will spend the majority of its time in deep space. However, it begins its mission in geostationary orbit (GEO). From that point, our spacecraft will travel across the solar system, gaining mechanical energy through planetary gravity assists. As of right now, our plan is to use a single Oberth maneuver at Jupiter. After this, the spacecraft will dive towards the sun, reaching a perihelion of 0.046 Astronomical Units (AU). At perihelion, the spacecraft will conduct another Oberth maneuver directed towards 650 AU. The estimated time for the mission itself will be 5 years for pre-perihelion maneuvering, and 30 years for the post-perihelion trajectory.

B. Power Loads

Table 1 Spacecraft Power Requirements by Subsystem

Subsystem	Power Requirement (W)
Telemetry System	15
Payload	10
ADCS	28
Propulsion System	20
Onboard Computer	25
Thermal Heaters	12

II. EPS Architecture Selection

The EPS will consist of a Multi-Hundred Watt Radioisotope Thermoelectric Generator (MHW-RTG) with the ability to produce 157 Watts at beginning of life[1], with a Nickel-Hydrogen Battery system. The power will be distributed via Direct Energy Transfer (DET) with a Power Conditioning Unit for modular design. This will also include a regulated bus, with the selected bus voltage to be 28V DC[2]. Shunt regulators and mechanical switching will be used to manage this voltage throughout the subsystems. Peak power tracking will not be needed due to the predictable continuous power produced by the MHW-RTG. As such, it is assumed that the peak power is equivalent to the average for each subsystem. It will have various modes, such as Propulsion mode during travel, Nominal science mode at the destination, a Downlink mode to transfer data, and a Safe mode for load shedding and focused essential subsystem upkeep. The EPS will have resistance to single point failure by incorporating protected switching and modular power distribution.

A. Key Design Drivers

The Key Design for the EPS lies in the mission duration and the distance it holds from the sun. The Solar Gravitational Lense is required to be placed at its closest 650 AU from the sun. A solar array will be rendered unusable soon after the spacecraft leaves the solar system, which must be accounted for. The full duration of the mission will exceed 35+ years. Due to these constraints, the main selection of power will come from Radioisotope Power Systems, otherwise known as Radioisotope Thermoelectric Generators (RTG)[3]. This is chosen because of its proven ability to survive for a long period of time and its continuous power generation for a low-power spacecraft. The isotope chosen for the generator will be Pu-238, with a half-life of 87.7 years, and the mechanical efficiency reduced effectively by .8% per year, the overall efficiency will be 1.6%. This means the spacecraft will maintain 78 Watts throughout the mission time of 35 years, with peak power supported through the travel time of 25 years. These constraints are also key to the

selection of the Nickel-Hydrogen batteries, with a vast cycle time of 15000+[4]. Though the MHW-RTG will produce continuous DC power to all systems, a battery design will be important to buffer transient loads and provide stability to power during load switching stages. DET is chosen due to the steady power output of the MHW-RTG, as well as having more reliability and less complexity in design.

III. Solar Array Sizing

Due to the nature of our mission, we found that it would be inefficient to incorporate a solar array into our design. Much of the mission will be in deep-space, where the power of the sun is negligible.

IV. Battery Sizing

A. Design Choice

DC converters, mechanical switching systems, and shunt regulators will be used to control bus voltage. We will use 28 V DC as the optimum regulatory voltage throughout the bus, as has been historically proven in NASA missions such as Voyager and Pioneer. The spacecraft will use a modular storage design comprising a Nickel-Hydrogen (NiH₂) cell system to buffer transient loads and provide power stability during load-switching events. There will be 23 cells in series, as NiH₂ is 1.25 V/cell. Conservative depth-of-discharge (DOD) and cycle times are chosen via NASA design standards. In this case, to reach 15,000+ cycles, the battery will be limited to 40 percent DOD maximum. This will include a built-in power conditioning unit (PCU) to provide control electronics for the EPS. Shunt and DC regulators previously mentioned will be implemented to control output voltage and load isolation. Protective elements such as fuses and relays will be installed as well.

B. Calculations

1. General Power Assumptions

$V_{\text{bus}} = 28\text{V}$
 $\text{DOD}_{\text{max}} = 0.40$
 Normal Load: $P = 78\text{W}$
 $\text{NiH}_2 = 1.25 \text{ V/cell}$
 Specific Energy of NiH_2 (from table) = $SE = 31.59\text{Wh/kg}$

2. Contingency Assumptions

Contingency ride-through duration $t = 30 \text{ mins} = 0.5 \text{ hrs}$ (since no eclipse)
 Peak contingency load: $P = 98 \text{ W}$
 Margin: $M = 1.20$
 Discharge + conditioning efficiency: $\eta = 0.90$

3. Energy Required During Contingency

$$E_{\text{load}} = P_{\text{peak}} \cdot t = (98 \text{ W})(0.50 \text{ hr}) = 49.0 \text{ Wh} \quad (1)$$

$$E_{\text{req}} = \frac{E_{\text{load}} \cdot M}{\eta} = \frac{(49.0)(1.20)}{0.90} = 65.3 \text{ Wh} \quad (2)$$

4. Required amp-hour capacity

$$E_{\text{usable}} = V_{\text{bus}} \cdot C_{\text{Ah}} \cdot \text{DOD}_{\text{max}} \quad (3)$$

$$C_{\text{Ah}} = \frac{65.3}{(28)(0.40)} = 5.83 \text{ Ah} \quad (4)$$

5. Total Stored Energy

$$E_{\text{stored}} = V_{\text{bus}} \cdot C_{\text{Ah}} = (28)(5.83) = 163 \text{ Wh} \quad (5)$$

6. Number of Cells in Series

$$N_s = \left\lceil \frac{V_{\text{bus}}}{V_{\text{cell}}} \right\rceil = \left\lceil \frac{28}{1.25} \right\rceil = 23 \text{ cells} \quad (6)$$

$$V_{\text{string}} = 23 \cdot 1.25 = 28.75 \text{ V} \quad (7)$$

7. Battery Mass Estimate

$$m_{\text{batt}} = \frac{E_{\text{stored}}}{SE} = \frac{163}{31.6} = 5.16 \text{ kg} \quad (8)$$

8. Final Calculated Values

Required amp-hour capacity: $C_{\text{Ah}} = 5.83 \text{ Ah}$

Total stored energy (Wh): $E_{\text{stored}} = 163 \text{ Wh}$

number of cells in series: 23

estimated battery mass = 5.2 kg

V. Power Balance and Energy Margin

To demonstrate the spacecraft EPS satisfies the energy balance requirement over a representative operational cycle, we evaluate the inequality:

$$E_{\text{in}} - E_{\text{out}} \geq 0$$

over a 24-hour schedule consisting of a peak downlink segment and nominal cruise operations.

A. Input power model

Electrical generation is provided by an RTG with beginning-of-life (BoL) output power $P_{\text{RTG}, \text{BoL}} = 157 \text{ W}$. RTG power is degraded at 1.6% per year and modeled as:

$$P_{\text{in}}(t) = 157(1 - 0.016)^{t_{\text{yr}}} \text{ W}$$

where t_{yr} is mission elapsed time in years.

B. Load schedule (24 hr cycle)

The power demand profile is:

- Downlink (peak): $P_{\text{out}} = 98 \text{ W}$ for 0.5 hr
- Nominal cruise: $P_{\text{out}} = 78 \text{ W}$ for 23.5 hr

Thus, the 24-hour energy out is:

$$E_{\text{out}} = 98(0.5) + 78(23.5) = 1882 \text{ Wh}$$

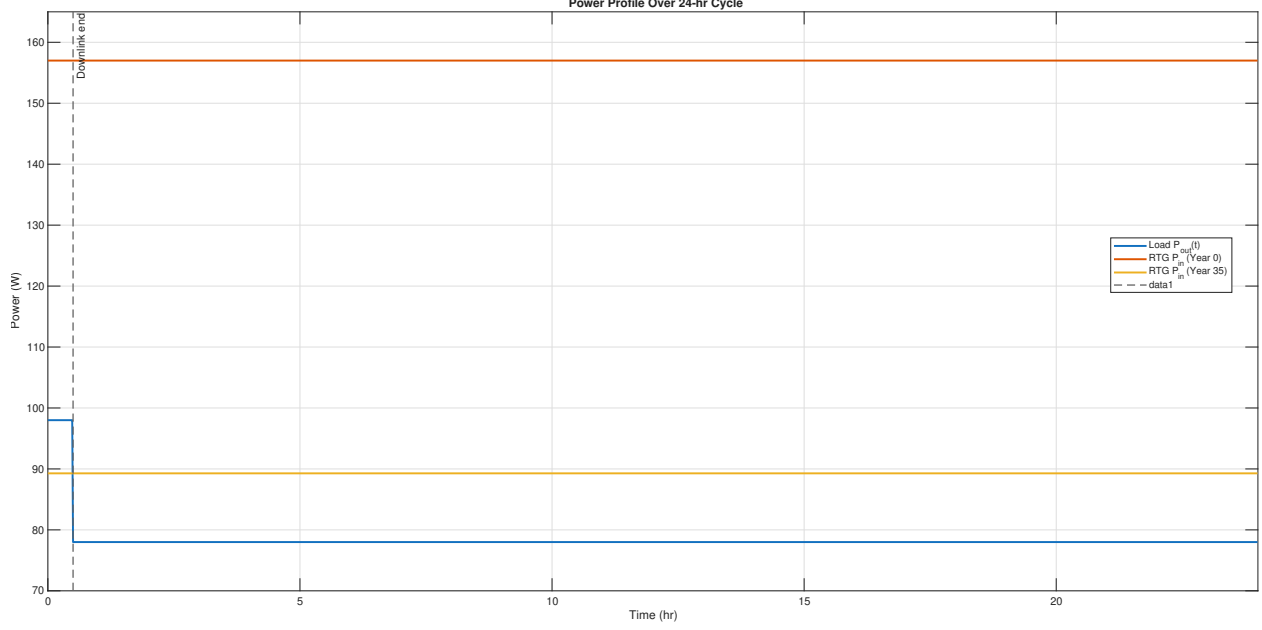


Fig. 1 Power profile over the 24-hr cycle. (Appendix VII.A)

C. Energy margin

At BoL, the 24-hour energy in is:

$$E_{in,BoL} = 157(24) = 3768 \text{ Wh}$$

so the energy margin is:

$$E_{margin,BoL} = E_{in,BoL} - E_{out} = 1886 \text{ Wh} \geq 0$$

At an end-of-life (EoL) check of 35 years, $P_{in} \approx 157(0.984)^{35} \approx 89 \text{ W}$, giving:

$$E_{in,EoL} \approx 89(24) = 2136 \text{ Wh}$$

$$E_{margin,EoL} = 2136 - 1882 = 254 \text{ Wh} \geq 0$$

Therefore, the EPS satisfies $E_{in} - E_{out} \geq 0$ for the full 24-hour operational cycle at both BoL and EoL point that represents a 30-40 year long mission.

D. Eclipse energy balance

The spacecraft is RTG-powered; therefore, electrical generation is independent of solar illumination and eclipse does not reduce P_{in} . Eclipse effects are limited to potential increases in electrical load (e.g., thermal control/heaters), which would reduce energy margin. The worst-case power event (downlink at 98 W) is evaluated at EoL and still satisfies $E_{in} - E_{out} \geq 0$ over the 24-hr cycle.

E. Peak power events

The worst-case electrical event considered is a downlink segment at 98 W for 0.5 hr. At EoL, $P_{in} \approx 89 \text{ W}$ so the instantaneous power deficit is approximately 9 W. Over 0.5 hr this corresponds to $\Delta E \approx 9(0.5)/0.90 \approx 5 \text{ Wh}$ of battery discharge.

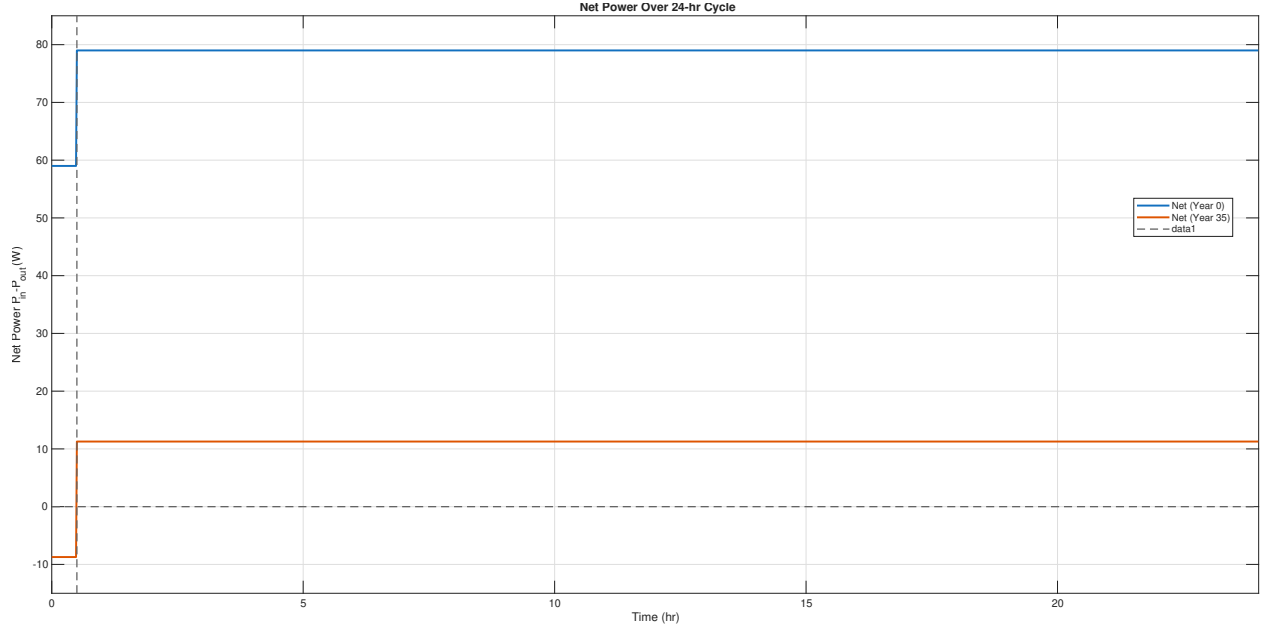


Fig. 2 Positive = Charging, Negative = Discharge. (Appendix VII.A)

F. Battery margin

The battery is sized with $E_{battery} = 163 \text{ Wh}$ and allowable $DOD_{max} = 40\%$. Using $\eta = 0.90$ for discharge/conditioning, the EoL downlink event draws approximately 5 Wh , corresponding to a maximum depth of discharge of $DOD_{used} \approx 5/163 \approx 3\% \ll 40\%$. Therefore, the battery provides large margin for transient peak events and contingency operations, while the RTG supplies the long-term average power required for multi-decade cruise.

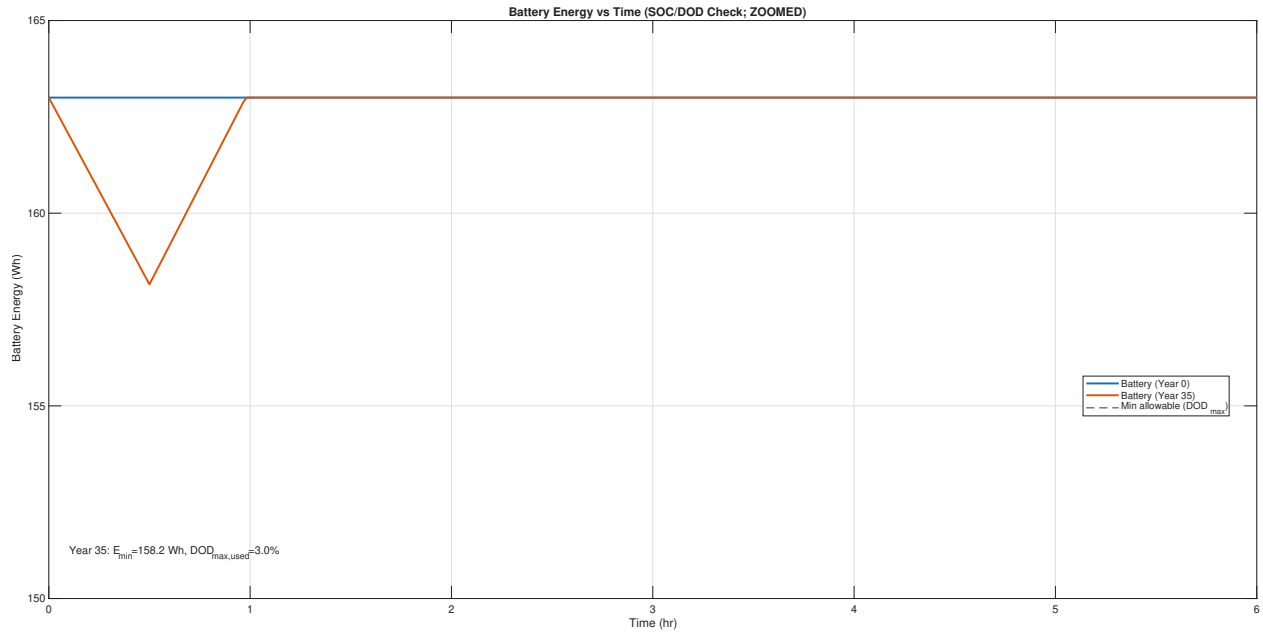


Fig. 3 Zoomed to show the dip better. (Appendix VII.A)

VI. Design Discussion and Trades

A. Design Discussion and Trades

The electrical power subsystem (EPS) design is driven primarily by the mission environment and the requirement for continuous autonomous operation in deep space. Unlike LEO spacecraft, this mission is not solar-powered due to the extremely low solar flux at the operational distance. As a result, the baseline architecture relies on a radioisotope thermoelectric generator (RTG) as the primary power source, supported by a rechargeable battery for transient loads and fault recovery.

1. Dominant EPS Drivers

The dominant EPS drivers for this mission are continuous survivability, reliability, and long-duration performance degradation. The RTG provides steady output power but experiences degradation over time, which must be accounted for in the long-term energy balance. In addition, certain mission modes such as downlink or propulsion operation may impose short-duration peak loads that exceed the RTG's instantaneous capability. For this reason, a battery is required even in eclipse-free operation, primarily for load leveling and contingency support.

Another key driver is fault tolerance. Due to the large communication latency and potential blackout intervals, the spacecraft must autonomously detect abnormal power conditions and perform fault detection, isolation, and recovery (FDIR). This would require the use of a regulated bus architecture with a power control unit (PCU) and a power distribution unit (PDU) capable of switching loads and shedding non-critical subsystems during emergencies.

2. Regulated Bus vs. Unregulated Bus

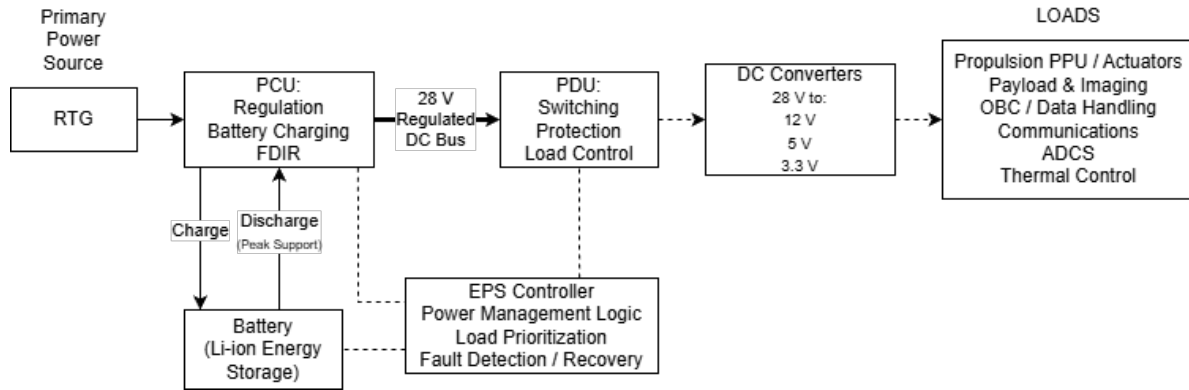


Fig. 4 Electrical Power Subsystem (EPS) architecture for the SGL mission, showing RTG-based power generation, regulated 28 V DC bus, battery energy storage, and distributed spacecraft loads.

A regulated DC bus architecture was selected for this mission because it simplifies subsystem power interfaces and improves stability under variable load conditions. In an unregulated bus system, bus voltage may vary significantly depending on battery state of charge and instantaneous load, increasing the complexity of downstream subsystem power electronics. A regulated 28 V bus is consistent with common spacecraft EPS practice and provides an efficient distribution voltage while remaining compatible with DC/DC conversion to lower subsystem voltages.

3. Battery Role and Trade Considerations

Although the spacecraft may not experience orbital eclipse, the battery is still required to handle peak-power events, including downlink operations and possible propulsion or attitude control excursions. The battery also provides a safety role by allowing continued operation of the spacecraft computer and communication system if the RTG output temporarily drops below required levels or during PCU switching events. The maximum depth of discharge (DOD) is constrained to improve battery cycle life and reduce degradation over the mission duration.

A trade exists between larger battery capacity and spacecraft mass. Increasing battery capacity improves margin and

survivability but directly increases dry mass, which negatively impacts propulsion performance and mission feasibility. The baseline battery design therefore prioritizes sufficient capacity for worst-case operational modes while minimizing mass.

4. Power Dissipation and Shunt Regulation

Since the RTG generates power continuously, excess power must be managed when the spacecraft is operating in low-power modes or when the battery is fully charged. This requires the inclusion of a voltage regulator module (or shunt module) or dump load capability within the PCU. Without this feature, bus voltage could rise beyond acceptable limits during low-load conditions. The shunt system provides controlled dissipation of unused power to maintain bus stability and protect downstream loads. A thorough analysis of rms, peak and safety margin for voltage and current would be necessary to determine the inductor, capacitor, and mosfet configuration for each specific load.

5. BOL vs. EOL Sensitivity

The system must be designed to meet power requirements not only at beginning-of-life (BOL), but also at end-of-life (EOL), where RTG power output is reduced due to degradation. A sensitivity analysis shows that decreasing available input power reduces operational margin and may require reduction in duty cycles for high-power modes such as downlink. This could require a conservative design approach in which nominal operational loads are kept significantly below the available RTG power, leaving margin for degradation and unforeseen contingencies.

6. Comparison to Heritage Systems

The selected EPS architecture is comparable to heritage deep-space missions that rely on radioisotope power sources. Similar to missions such as Voyager, Cassini, and New Horizons, the use of an RTG provides long-duration continuous power independent of solar flux. However, the relatively small power budget in this mission imposes strict load management requirements, motivating an autonomous power management system capable of load shedding and prioritized operation.

7. Key Risks and Mitigations

The primary EPS risks are power shortfall at EOL, thermal waste heat management from the RTG, and battery degradation. These risks are mitigated by conservative power margins, regulated bus control with FDIR logic, and limiting battery depth of discharge. Additional risk reduction can be achieved by validating subsystem duty cycles early in the mission design and ensuring the spacecraft can enter a low-power safe mode while maintaining attitude control and communication capability.

VII. Conclusion

This technical memorandum presented the preliminary design for the EPS design for the Solar Gravitational Lense mission. With the capability of sustaining continuous operation over a 35+ year duration within deep space. The final EPS architecture involves a Multi-Hundred Watt Radioisotope Thermoelectric Generator, a regulated 28 Volt DET bus, a modular power distribution and Nickerl-Hydrogen battery storage. This design is made with simplicity, operational stability, and reliability in mind.

No solar arrays are required due to the extreme distance from the Sun. Battery sizing results in a required capacity of 5.83 Ah (163 Wh total), implemented with 23 NiH₂ cells in series and an estimated mass of approximately 5.2 kg. This configuration provides substantial margin for transient peak loads and contingency operations, with only a small fraction of allowable depth-of-discharge utilized during worst-case events.

Key assumptions include an RTG power degradation rate of 1.6% per year and stable long-term thermal and electrical performance of both the RTG and battery system. The risks are mainly associated with the performance of the RTG over the long duration of the mission, with a possibility for variable degradation over the multiple decades, along with possible increases in power demand from the subsystems later in the mission. Though given proven heritage in electronics and the analytical results, the positive energy margin confirms EPS meets all power requirements while maintaining high reliability.

Appendix

A. MATLAB

1. Part 5

```
%% TM2 part 5
clear; clc;

% Given
P_RTG_BOL = 157;           % W
deg_per_yr = 0.016;        % 1.6%/yr
eta_chg    = 0.90;
eta_dis    = 0.90;

Ebat_cap_Wh = 163;         % Wh
DOD_max     = 0.40;

% 24-hr schedule
P_downlink = 98;           % W
t_downlink = 0.5;          % hr
P_nominal   = 78;          % W
T_hr        = 24;          % hr

% Sim years
years_list = [0 35];

% Time grid
dt_min = 1;                % minutes
dt_hr   = dt_min/60;
t = 0:dt_hr:T_hr;
N = numel(t);

% Load profile
P_load = P_nominal * ones(size(t));
P_load(t < t_downlink) = P_downlink;

% Pre
Ebat_hist = zeros(numel(years_list), N);
Pin_hist   = zeros(numel(years_list), N);
Pnet_hist  = zeros(numel(years_list), N);

for iy = 1:numel(years_list)
    yrs = years_list(iy);

    % RTG power at year yrs
    P_in = P_RTG_BOL * (1 - deg_per_yr)^yrs;
    Pin_hist(iy,:) = P_in;

    % Initialize battery full
    Ebat = Ebat_cap_Wh;
    Ebat_hist(iy,1) = Ebat_cap_Wh;

    % Integrals
```

```

Ein_Wh = 0; Eout_Wh = 0;

for k = 1:N-1
    Ein_Wh = Ein_Wh + P_in * dt_hr;
    Eout_Wh = Eout_Wh + P_load(k) * dt_hr;

    P_net = P_in - P_load(k); % + = charge surplus, - = deficit
    Pnet_hist(iy,k) = P_net;

    if P_net >= 0
        Ebat = min(Ebat_cap_Wh, Ebat + (P_net * dt_hr * eta_chg));
    else
        Ebat = Ebat + (P_net * dt_hr / eta_dis); % P_net negative
    end

    Ebat_hist(iy,k+1) = Ebat;
end

% Metrics
E_margin_Wh = Ein_Wh - Eout_Wh;
Ebat_min = min(Ebat_hist(iy,:));
DOD_used = (Ebat_cap_Wh - Ebat_min)/Ebat_cap_Wh;

fprintf('\n===== Year %g =====\n', yrs);
fprintf('RTG input power: %.2f W\n', P_in);
fprintf('Energy IN (24 hr): %.2f Wh\n', Ein_Wh);
fprintf('Energy OUT (24 hr): %.2f Wh\n', Eout_Wh);
fprintf('Energy margin (IN-OUT): %.2f Wh\n', E_margin_Wh);
fprintf('Min battery energy: %.2f Wh\n', Ebat_min);
fprintf('Max DOD used: %.2f %% (limit %.2f %%)\n', 100*DOD_used, 100*
    DOD_max);
end

%% Plot 1: Power profile (zoomed)
figure;
plot(t, P_load, 'LineWidth', 1.8); hold on;
plot(t, Pin_hist(1,:), 'LineWidth', 1.8);
plot(t, Pin_hist(2,:), 'LineWidth', 1.8);
xlabel('Time (hr)'); ylabel('Power (W)');
legend('Load P_{out}(t)', 'RTG P_{in} (Year 0)', 'RTG P_{in} (Year 35)', '
    Location','best');
title('Power Profile Over 24-hr Cycle');
ylim([70 165]); xlim([0 24]); grid on;
xline(t_downlink, '--', 'Downlink end', 'LineWidth', 1.2);
figure(1);
set(gcf, 'Renderer', 'painters');
print(gcf, '-depsc2', '-r600', 'power_profile.eps');

%% Plot 2: Net power
figure;
plot(t, (Pin_hist(1,:) - P_load), 'LineWidth', 1.8); hold on;
plot(t, (Pin_hist(2,:) - P_load), 'LineWidth', 1.8);
yline(0, '--', 'LineWidth', 1.2);
xlabel('Time (hr)'); ylabel('Net Power P_{in}-P_{out} (W)');

```

```

legend('Net (Year 0)', 'Net (Year 35)', 'Location','best');
title('Net Power Over 24-hr Cycle');
xlim([0 24]); grid on;
xline(t_downlink,'--','LineWidth',1.2);
ylim([-15 85]);
figure(2);
set(gcf,'Renderer','painters');
print(gcf,'-depsc2','-r600','net_power.eps');

%% Plot 3: Battery energy (zoomed)
figure;
plot(t, Ebat_hist(1,:), 'LineWidth', 1.8); hold on;
plot(t, Ebat_hist(2,:), 'LineWidth', 1.8);
yline(Ebat_cap_Wh*(1-DOD_max), '--', 'LineWidth', 1.2);
xlabel('Time (hr)'); ylabel('Battery Energy (Wh)');
legend('Battery (Year 0)', 'Battery (Year 35)', 'Min allowable (DOD_{max})', '
Location','best');
title('Battery Energy vs Time (SOC/DOD Check; ZOOMED)');
xlim([0 6]); % zoom
ylim([150 165]); % zoom
grid on;
Ebat_min35 = min(Ebat_hist(2,:));
DOD_used35 = (Ebat_cap_Wh - Ebat_min35)/Ebat_cap_Wh;
txt = sprintf('Year 35: E_{min}=%.1f Wh, DOD_{max,used}=%.1f%%',Ebat_min35
,100*DOD_used35);
text(0.1,151.2,txt,'FontSize',10);
figure(3);
set(gcf,'Renderer','painters');
print(gcf,'-depsc2','-r600','battery_energy.eps');

```

Nomenclature

A_{craft}	=	Total external surface area of the spacecraft bus
AU	=	Astronomical Unit
BoL	=	Beginning of Life
C_{Ah}	=	Battery amp-hour capacity (Ah)
DOD	=	Depth of Discharge
DOD_{max}	=	Maximum allowable depth of discharge
DET	=	Direct Energy Transfer
E_{battery}	=	Total battery stored energy (Wh)
E_{in}	=	Electrical energy generated over time interval (Wh)
E_{out}	=	Electrical energy consumed over time interval (Wh)
E_{load}	=	Energy required by loads during contingency period (Wh)
E_{req}	=	Required battery energy including margin and efficiency losses (Wh)
E_{stored}	=	Total stored battery energy (Wh)
EoL	=	End of Life
η	=	Electrical efficiency (charge/discharge and conditioning)
M	=	Energy margin factor
m_{batt}	=	Battery mass (kg)
N_s	=	Number of battery cells in series
NiH_2	=	Nickel-Hydrogen battery chemistry
OBC	=	On-Board Computer
P	=	Electrical power (W)
P_{in}	=	Electrical power generated by RTG (W)
P_{out}	=	Electrical power consumed by spacecraft loads (W)
P_{peak}	=	Peak contingency power demand (W)
$P_{RTG,BoL}$	=	RTG output power at beginning of life (W)
PCU	=	Power Conditioning Unit
$Pu-238$	=	Plutonium-238 radioisotope fuel
RTG	=	Radioisotope Thermoelectric Generator
SE	=	Battery specific energy (Wh/kg)
SGL	=	Solar Gravitational Lens
t	=	Time
t_{yr}	=	Mission elapsed time in years
$TT\&C$	=	Telemetry, Tracking, and Control
V_{bus}	=	Spacecraft power bus voltage (V)
V_{cell}	=	Single battery cell voltage (V)
V_{string}	=	Battery string voltage (V)

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